KAIZEN MATHS

Lesson starter

Product, Quotient and Chain Rule

Applications

Name _____

S1

Find the tangent to $y = 2x^2 + 4x + 2$ at $x = 1$.

S2

Find turning points of $y = 2x^3 - 9x^2 + 12x - 4$.

S3

Find the tangent to $y = 3x^2 + 2x + 2$ at $x = 2$.

S4

Find the tangent to $y = 2x^2 + 4x + 2$ at $x = 2$.

Show your working

KAIZEN MATHS

Lesson starter

Product, Quotient and Chain Rule

Applications

Name _____

S1

Find the tangent to $y = 2x^2 + 4x + 2$ at $x = 1$.

S2

Find turning points of $y = 2x^3 - 9x^2 + 12x - 4$.

S3

Find the tangent to $y = 3x^2 + 2x + 2$ at $x = 2$.

S4

Find the tangent to $y = 2x^2 + 4x + 2$ at $x = 2$.

ALG-1258-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Product, Quotient and Chain Rule

Applications

Name _____

E1

Find the tangent to $y = 1x^2 + 3x + 2$ at $x = 1$.

KAIZEN MATHS

Exit ticket

Product, Quotient and Chain Rule

Applications

Name _____

E1

Find the tangent to $y = 1x^2 + 3x + 2$ at $x = 1$.

ALG-1258-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Product, Quotient and Chain Rule

Applications

Name _____

E1

Find the tangent to $y = 1x^2 + 3x + 2$ at $x = 1$.

ALG-1258-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Product, Quotient and Chain Rule

Applications

Name _____

E1

Find the tangent to $y = 1x^2 + 3x + 2$ at $x = 1$.

ALG-1258-L4 · E

Show your working

Teacher answers & working

Product, Quotient and Chain Rule · Applications

ALG-1258-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Find the tangent to $y = 2x^2 + 4x + 2$ at $x = 1$.

Answer:

$y = 8x + 0$

Working:

$y' = 2 \cdot 2x^{2-1} + 4 + 0$

$y' = 4x + 4$

$m = 4(1) + 4$

$m = 4 + 4 = 8$

$y_1 = 2(1)^2 + 4(1) + 2$

$y_1 = 2 + 4 + 2 = 8$

$y - y_1 = m(x - x_1)$

$y - 8 = 8(x - 1)$

$y - 8 = 8x - 8$

$y = 8x - 8 + 8$

$y = 8x + 0$

S2

Find turning points of $y = 2x^3 - 9x^2 + 12x - 4$.

Answer:

Maximum at $(1,1)$; minimum at $(2,0)$.

Working:

$y' = 2 \cdot 3x^{3-1} - 9 \cdot 2x^{2-1} + 12$

$y' = 6x^2 - 18x + 12$

Set the gradient to zero: $6x^2 - 18x + 12 = 0$.

Divide by 6: $x^2 - 3x + 2 = 0$.

Factor: $(x - 1)(x - 2) = 0$.

$x - 1 = 0$ or $x - 2 = 0$.

$x = 1$ or $x = 2$.

$y(1) = 2(1)^3 - 9(1)^2 + 12(1) - 4 = 1$

$y(2) = 2(2)^3 - 9(2)^2 + 12(2) - 4 = 0$

$y'' = 12x - 18$

$y''(1) = 12(1) - 18 = -6 < 0$: maximum.

$y''(2) = 12(2) - 18 = 6 > 0$: minimum.

Maximum at $(1,1)$; minimum at $(2,0)$.

Teacher answers & working

Product, Quotient and Chain Rule · Applications

ALG-1258-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Find the tangent to $y = 3x^2 + 2x + 2$ at $x = 2$.

Answer:

$y = 14x - 10$

Working:

$y' = 3 \cdot 2x^{2-1} + 2 + 0$

$y' = 6x + 2$

$m = 6(2) + 2$

$m = 12 + 2 = 14$

$y_1 = 3(2)^2 + 2(2) + 2$

$y_1 = 12 + 4 + 2 = 18$

$y - y_1 = m(x - x_1)$

$y - 18 = 14(x - 2)$

$y - 18 = 14x - 28$

$y = 14x - 28 + 18$

$y = 14x - 10$

S4

Find the tangent to $y = 2x^2 + 4x + 2$ at $x = 2$.

Answer:

$y = 12x - 6$

Working:

$y' = 2 \cdot 2x^{2-1} + 4 + 0$

$y' = 4x + 4$

$m = 4(2) + 4$

$m = 8 + 4 = 12$

$y_1 = 2(2)^2 + 4(2) + 2$

$y_1 = 8 + 8 + 2 = 18$

$y - y_1 = m(x - x_1)$

$y - 18 = 12(x - 2)$

$y - 18 = 12x - 24$

$y = 12x - 24 + 18$

$y = 12x - 6$

Teacher answers & working

Product, Quotient and Chain Rule · Applications

ALG-1258-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Find the tangent to $y = 1x^2 + 3x + 2$ at $x = 1$.

Answer:

$y = 5x + 1$

Working:

$y' = 1 \cdot 2x^{2-1} + 3 + 0$

$y' = 2x + 3$

$m = 2(1) + 3$

$m = 2 + 3 = 5$

$y_1 = 1(1)^2 + 3(1) + 2$

$y_1 = 1 + 3 + 2 = 6$

$y - y_1 = m(x - x_1)$

$y - 6 = 5(x - 1)$

$y - 6 = 5x - 5$

$y = 5x - 5 + 6$

$y = 5x + 1$